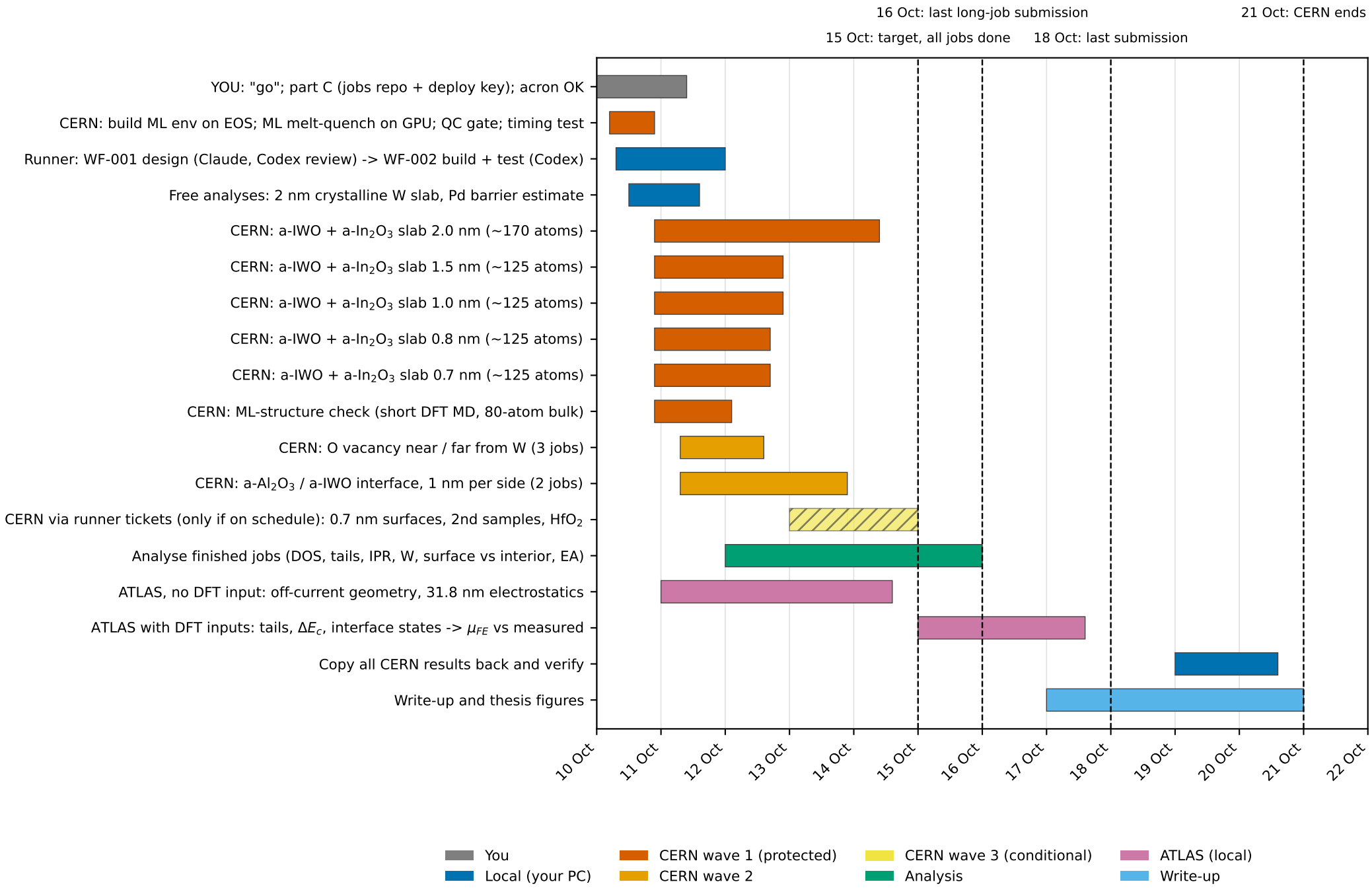
IWO project: CERN DFT + ATLAS timeline, 10-21 October 2026

Durations are estimates from past CERN timings (80-atom cell: ~2.5 h per SCF); the timing test on 10 Oct replaces them with measured values.
No new measurements: inputs are the existing curves, the published paper, CERN (until 21 Oct; ML and DFT both run there) and local Silvaco ATLAS.



Jobs: what is built and run

Amorphous slab series (ML-built on CERN GPUs, DFT-relaxed; 1 sample each; IWO + In2O3 twin = 10 structures)

Slab	In-plane cell	Atoms (+H)	W content	Twin	Est. DFT time	Priority
2.0 nm	1.05 x 1.05 nm	~170	1 W = 1.5 % (2 % needs ~250 atoms)	a-In2O3	2-4 days	1 (protected)
1.5 nm	1.05 x 1.05 nm	~125	1 W = 2.0 %	a-In2O3	1-2 days	1
1.0 nm	1.29 x 1.29 nm	~125	1 W = 2.0 %	a-In2O3	1-2 days	1
0.8 nm	1.44 x 1.44 nm	~125	1 W = 2.0 %	a-In2O3	1-2 days	1
0.7 nm	1.54 x 1.54 nm	~125	1 W = 2.0 %	a-In2O3	1-2 days	1

Other jobs

Job	Structure	Question	Est. time	Priority
ML-structure check	80-atom a-bulk, short DFT MD	Is the ML-built structure stable and right under DFT?	~1 day	2
O vacancy near / far W	79-atom crystal cells (3 jobs)	Does W raise the vacancy formation energy? By how much?	~1 day	3
Oxide interface	a-Al2O3 / a-IWO, 1 nm/side (2 jobs)	Band offset, interface states (-> D_it, Q_f in TCAD)	2-3 days	4
Conditional extras	0.7 nm surfaces; 2nd samples; HfO2	Surface chemistry; statistics; the paper's interface	1-2 days	5 (if on time)
Free (no CERN)	existing outputs	2 nm crystal W slab; Pd barrier = WF(Pd) - EA(IWO)	hours	-

How the jobs depend on each other, what is computed, and the rules

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DEPENDENCIES (serial)
  one DAG per structure, all on CERN (laptop can be off):
  ML (GPU) -> QC gate -> DFT relax (<=20 h segments) -> SCF -> DOS/IPR
  ML-structure check must agree before the slab results are called final
  ATLAS runs with DFT inputs wait for the slab analysis

AUTOMATIC RUNNER (from ~12 Oct)
  ticket -> you merge = approve -> acron on lxplus -> DAG
  -> results pushed to the private iwo-dft-jobs repo

PARALLEL
  all CERN jobs (no job needs another job's result)
  local analysis and ATLAS run while CERN computes

COMPUTED ON EVERY SLAB
  1  structure: coordination, defect sites, W site, surface OH
  2  band gap vs thickness (confinement), HSE-corrected
  3  band tails: width below the conduction band (vs TCAD W_TA = 40 meV)
  4  localization (IPR): does W keep states delocalized when thin?
  5  W: donor level, charge, Fermi level
  6  surface vs interior DOS (tests the N_s + N_b t trap split)
  7  electron affinity vs thickness (Delta E_c law for TCAD)

MOBILITY
  not computed by DFT; DFT gives tails, m*, IPR, n_free/n_total;
  ATLAS turns them into mu_FE, which is compared with
  the measured 12 / 11 / 50 cm2/Vs (2.0 / 6.3 / 13.2 nm)

DEADLINES
  10 Oct  everything built and wave 1 submitted
  11 Oct  wave 2 submitted
  13-14 Oct  wave 3 only if wave 1 is on schedule
  15 Oct  target: all jobs finished
  16 Oct  last submission of a job > 2 days (fallback)
  18 Oct  last submission of any job
  19-20 Oct  all results copied back and verified
  21 Oct  CERN access ends (buffer, nothing running)

CUT ORDER (if CERN is busy or slipping)
  first cut: extras -> interface -> In2O3 twins at 1.5 / 0.8 nm
  protected: all five a-IWO slabs, 2 nm In2O3 twin, ML check

CERN SAFETY (lessons from Sept-Oct)
  restart files written regularly;
  snapshot_jobs.sh copies outputs every 15 min
  (running jobs lose EOS access after ~25 h)
  never condor_release a job that ran > 1 h:
  resubmit from the snapshot instead
  you log in; passwords / 2FA are never handled by an agent

CAVEATS TO STATE AT THE DEFENCE
  PBE gaps are underestimated: trends and widths, not absolute gaps
  small cells, fast quench: more defective than real films
  1 sample per structure: indicative statistics
  slab surfaces are H/OH in vacuum, not the real oxide / air
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